
CS 686: Large Language Models: Core Concepts to Custom Applications

Course Code	CS 686
Course Name	Large Language Models: Core Concepts to Custom Applications
L-T-P-C	3-0-2-4
Prerequisites	Introductory course in machine-learning/AI; Programming in Python
Intended for	BTech 3rd/4th year; MTech; MSc; Ph.D
Approval	54th BoA (May 2024)

Course Contents

- Introduction to NLP and LLMs (9 Hours)
 - Fundamentals of NLP and its Evolution
 - Language Model Basics: n-grams to Neural Networks
 - Word embeddings
 - Introduction to Transformer Architectures and Attention Mechanisms
 - Introduction to Generative AI
 - Overview of Prominent LLMs (GPT, BERT, Llama, Mistral, etc.) and their Impact
- Deep Dive into LLM Architectures (12 Hours)
 - Selecting the right model and Generative AI responsibility
 - Comprehensive Study of Transformer Mechanism
 - Scaling Laws: Understanding computational demands and data efficiency
 - Hallucinations and Citations
 - Architectural Variants and Innovations in LLMs, including parameter-efficient fine-tuning models like LoRA, QLoRA
- Practical Applications and Customization of LLMs (15 Hours)
 - Text data cleaning, normalization, and tokenization
 - LLMs in Text Generation, Translation, and Summarization
 - Best Prompt Engineering Techniques
 - Fine-Tuning Strategies for Task-Specific Applications
 - Overview of Instruction Fine Tuning and Reinforcement Learning from Human Feedback (RLHF)
 - Model alignment techniques including Decision Process Optimization (DPO), Knowledge Transfer Optimization (KTO)
 - Retrieval-Augmented Generation (RAG)
 - Practical Session: Building LLM-based Applications
 - Building Chat Applications
 - Embeddings based Search Applications
 - Speech based LLMs
- Ethical Considerations and Societal Impacts (4.5 Hours)
 - Addressing Bias and Fairness in LLM Outputs
 - Ethical Use and Misuse of Generative AI
 - Future Societal Challenges and Opportunities
- Advanced Topics and Research Directions in LLMs (4.5 Hours)

- Emerging Trends in LLM Research
- Novel Applications and Future Technologies
- Open Problems and Discussion

Laboratory Modules

In addition to the lecture modules, this course will include take home assignment where students will apply their knowledge to develop and fine-tune LLM applications using Python and platforms such as Hugging Face Transformers.

Text Books

1. Yoav Goldberg, **Neural Network Methods for Natural Language Processing**, Morgan and Claypool, 2017.
2. Dan Jurafsky and James Martin, **Speech and Language Processing**, 3rd Edition.

References

1. Ian Goodfellow, Yoshua Bengio, Aaron Courville, **Deep Learning**, MIT Press, USA, 2016.
2. A collection of the latest research papers, technical blogs, and articles from top journals and conferences to stay updated with the advancements and debates in the domain of LLMs.
3. **Large Language Models in Practice** - This is a placeholder for a current and comprehensive resource focused on the practical and theoretical aspects of LLMs. Due to the rapidly evolving nature of this field, the latest resource available at the time of course launch should be selected.
4. Generative AI for beginners by, Microsoft, https://learn.microsoft.com/en-gb/collections/zpy7c8zmq6ky0z?WT.mc_id=academic-105485-koreyst